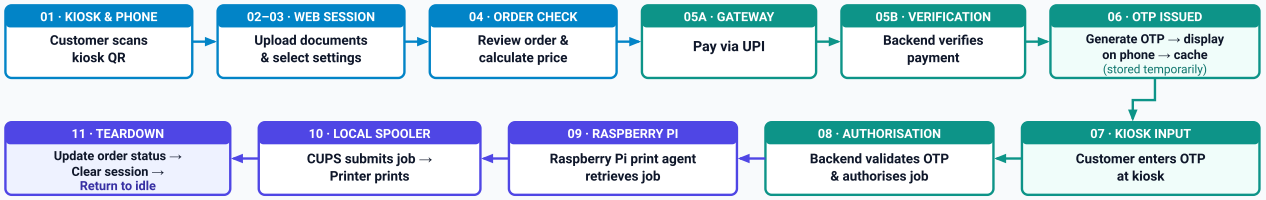
PrintBooth – OTP-Based Printing Workflow

END-TO-END CUSTOMER JOURNEY, TECHNICAL SEQUENCE & SECURITY SPECIFICATION

SYSTEM ARCHITECTURE

Technical Flow Overview

11-STAGE PIPELINE



■ Blue: Session & Document Setup (Steps 1–4)

■ Teal: UPI Payment & OTP Verification (Steps 5–8)

■ Indigo: Edge Execution & Reset (Steps 9–11)

Detailed Operational Workflow (Steps 1 – 6)

SESSION, PAYMENT & OTP ISSUANCE

01 **Customer scans the kiosk QR code** · KIOSK → PHONE

- The customer scans the QR code displayed on the PrintBooth kiosk using their phone.
- The website opens and establishes a session linked to that kiosk.

02 **Upload documents** · MOBILE WEB → BACKEND

- The customer uploads the documents they want to print.
- The system validates the uploaded files and confirms that they are ready.

03 **Select print settings** · CUSTOMER CONFIGURATION

The customer selects the required settings, such as:

- Number of copies
- Colour or black-and-white
- Single-sided or double-sided, if supported
- Page range and paper size, if supported

04 **Review the order** · PRICING & CONFIRMATION

- The customer reviews the uploaded documents, print settings, page count and total price.
- The customer confirms the order.

05 **Make payment** · UPI PAYMENT GATEWAY

- The system generates a UPI payment QR code through the payment gateway.
- The customer scans the QR code and completes the payment.
- The backend verifies the payment status directly with the payment gateway.

06 **Generate and display OTP** · BACKEND CACHE & MOBILE SCREEN

- Once payment is verified, the backend generates a unique OTP for the order.
- The OTP is displayed on the customer's phone screen; it is not sent by SMS.
- The OTP is temporarily stored in the cache with an expiration time.

Detailed Operational Workflow (Steps 7 – 11)

KIOSK VALIDATION, HARDWARE PRINT & RESET

07 **Enter OTP at the kiosk** · KIOSK INTERFACE → BACKEND CACHE

- The customer returns to the kiosk and enters the OTP using the kiosk's input interface.
- The backend validates the OTP against the cached record.
- The system checks that the OTP is correct, unexpired, unused and associated with the correct order and kiosk.

08 **Authorise the print job** · BACKEND STATE TRANSITION

- If validation succeeds, the backend authorises the print job.
- The OTP is marked as used so it cannot authorise another print job.
- If validation fails, the system displays an appropriate error and allows a limited number of retry attempts.

09 **Transfer the document to the Raspberry Pi** · PRINT AGENT & CUPS

- The Raspberry Pi's print agent securely retrieves the authorised document and print settings from the backend.
- The print agent submits the job to the local printing service, such as CUPS.

10 **Print the document** · HARDWARE EXECUTION & LIVE STATUS

- The printer processes the job and prints the document.
- The system updates the order status as printing progresses.
- The kiosk displays the success or failure status.

11 **Complete and reset** · SESSION CLEANUP & IDLE STATE

- The system marks the order as completed once printing is confirmed as far as the available printer status allows.
- Temporary session data is cleared according to the retention policy.
- The kiosk returns to the idle screen, ready for the next customer.

Important Security and Reliability Rules

MANDATORY SYSTEM SAFEGUARDS

CONTROL DOMAIN	ENFORCEMENT RULE & OPERATIONAL REQUIREMENT
Payment	Never authorise printing based only on the payment redirect or a screenshot. The backend must verify payment.
OTP	Use an expiry time, limit incorrect attempts and ensure each OTP can be consumed only once.
Order binding	Bind the OTP to its specific order and kiosk.
Duplicate prevention	Use a unique print-job ID and idempotent processing to prevent accidental duplicate prints.
Printer failures	If the printer is offline, out of paper or reports an error, update the order status and apply a defined retry or refund policy.
Session safety	A late payment or expired session must never trigger printing for the next customer.